

# Disproof of Conjecture 00000001564: a $k$ -dimensional cs-neighborly polytope can have only $2k < 2^{k+1}$ vertices

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## Abstract

Conjecture 00000001564 asserts that the minimal number of vertices of a  $k$ -dimensional cs-neighborly polytope is  $2^{k+1}$ , and that the extremal bodies are the Hanner polytopes. The first clause is false for every  $k \geq 1$ . The  $k$ -dimensional cross-polytope  $C_k^\Delta = \text{conv}\{\pm e_1, \dots, \pm e_k\}$  is centrally symmetric, has  $2k$  vertices, and is cs-neighborly in the strongest possible sense: every set of its vertices containing no antipodal pair is the vertex set of a face. Since  $2k < 2^{k+1}$ , the claimed minimum is wrong; the smallest cases are the square ( $4 < 8$ ) and the octahedron ( $6 < 16$ ). The counterexample works under every reading of “cs-neighborly” we know, including the statement’s own definition. We also show that the true minimum is  $2k$ , and that no Hanner polytope has  $2^{k+1}$  vertices. The refutation is formalized in Lean 4 (core library only) for  $k = 1, 2, 3, 4$ . The Chinese version of the statement asks about  $2k$ -dimensional polytopes. That claim holds for  $k = 1, 2$  and fails for every  $k \geq 3$ ; the smallest counterexample is the 6-dimensional cross-polytope ( $12 < 16$ ), also formalized in Lean.

## 1 The conjecture

*Definition:* A cs-neighborly polytope is centrally symmetric and has support points in every direction. *Conjecture:* The minimal number of vertices of a  $k$ -dimensional cs-neighborly polytope is  $2^{k+1}$ ; and the extremal bodies are the Hanner polytopes.

The Chinese version differs in one point: it speaks of a “ $2k$ -dimensional” polytope (the dimension is written as “two  $k$ ”), while the number of vertices is still  $2^{k+1}$ . We treat this reading separately in Section 5. The conjecture is a conjunction, so it is enough to refute the first clause:

$$\min\{f_0(P) : P \text{ a } k\text{-dimensional cs-neighborly polytope}\} = 2^{k+1} \quad (k \geq 1), \quad (1)$$

where  $f_0(P)$  is the number of vertices of  $P$ . A statement “the minimum is  $n$ ” contains the lower bound “every such  $P$  has  $f_0(P) \geq n$ ”. We refute this lower bound, which is the weakest part of (1).

**Readings of “cs-neighborly”.** The definition in the statement is unusual. “Having support points in every direction” holds for every nonempty compact set, so under the literal reading a cs-neighborly polytope is simply a centrally symmetric polytope. The standard notion is Grünbaum’s [1]. A centrally symmetric (cs) polytope  $P$  is *cs- $j$ -neighborly* if every set of at most  $j$  vertices that contains no pair of antipodal vertices  $\{v, -v\}$  is the vertex set of a face of  $P$ . “cs-neighborly” means cs-2-neighborly (any two non-antipodal vertices span an edge), or cs- $\lfloor k/2 \rfloor$ -neighborly for a  $k$ -polytope, depending on the author. We treat all these readings at once, via the strongest property:

(F) every nonempty set of vertices without an antipodal pair is the vertex set of a face.

Property (F) implies  $cs$ - $j$ -neighborliness for every  $j$ , and it implies the statement's literal definition. The cross-polytope has property (F) (Theorem 1), so it is a counterexample under every reading.

## 2 Definitions

A *polytope* is the convex hull  $P = \text{conv } V$  of a finite set  $V \subset \mathbb{R}^k$ . Its *dimension* is the dimension of its affine hull. A point  $v \in V$  is a *vertex* of  $P$  if some linear functional  $c$  attains its maximum over  $P$  only at  $v$ . Every vertex of  $P$  lies in  $V$ . If  $V$  is in *convex position* (every point of  $V$  is a vertex), then  $f_0(P) = |V|$ . For a functional  $c$ , the set  $F_c = \{x \in P : c \cdot x = \max_P c\}$  is a face of  $P$ . Its vertex set is  $F_c \cap V = \text{argmax}_{v \in V} c \cdot v$ , and every nonempty face of a polytope arises this way. So a set  $S \subseteq V$  is the vertex set of a nonempty face iff  $S = \text{argmax}_V c$  for some  $c$ .  $P$  is *centrally symmetric* if  $P - x_0 = -(P - x_0)$  for some center  $x_0$ . For  $x_0 = 0$  this means  $V = -V$  for the vertex set  $V$ .

## 3 The cross-polytope

**Theorem 1.** *Let  $k \geq 1$  and  $V_k = \{\pm e_1, \dots, \pm e_k\} \subset \mathbb{R}^k$ ,  $C_k^\Delta = \text{conv } V_k$ . Then:*

1.  $C_k^\Delta$  is  $k$ -dimensional and centrally symmetric (about 0);
2.  $V_k$  is in convex position, so  $f_0(C_k^\Delta) = 2k$ ;
3. every nonempty  $S \subseteq V_k$  containing no antipodal pair is the vertex set of a face, namely  $S = \text{argmax}_{V_k} c_S$  with  $c_S = \sum_{s \in S} s$ ;
4.  $C_k^\Delta$  has a support point in every direction.

*Proof.* (1)  $V_k = -V_k$ . The affine hull contains  $0 = \frac{1}{2}(e_1 + (-e_1))$  and every  $e_i$ , so it is  $\mathbb{R}^k$ . (2) For  $v \in V_k$  we have  $v \cdot v = 1$  and  $v \cdot w \in \{0, -1\}$  for  $w \in V_k \setminus \{v\}$ . So  $c = v$  is maximized over  $V_k$  only at  $v$ , and hence over  $C_k^\Delta$  only at  $v$ . (3) Since  $S$  contains at most one of  $e_i, -e_i$ , the  $i$ -th coordinate of  $c_S$  is  $+1$  if  $e_i \in S$ ,  $-1$  if  $-e_i \in S$ , and  $0$  otherwise. Hence  $c_S \cdot s = 1$  for  $s \in S$ . For  $w \in V_k \setminus S$  we get  $c_S \cdot w \in \{0, -1\}$ : it is  $-1$  if  $-w \in S$  and  $0$  otherwise. So  $\text{argmax}_{V_k} c_S = S$ . (4) Any functional attains its maximum on the compact set  $C_k^\Delta$ .  $\square$

## 4 The disproof

**Theorem 2.** *For every  $k \geq 1$  there is a  $k$ -dimensional centrally symmetric polytope with property (F) (hence  $cs$ - $j$ -neighborly for every  $j$ , and  $cs$ -neighborly in the sense of the statement) that has  $2k < 2^{k+1}$  vertices. Hence claim (1) is false for every  $k \geq 1$ , and so is the conjecture.*

*Proof.* Take  $C_k^\Delta$  (Theorem 1). We have  $2k \leq 2^k < 2^{k+1}$ , since  $2 \cdot 1 = 2^1$  and  $2(k+1) = 2k + 2 \leq 2^k + 2^k$ .  $\square$

The smallest instances are the segment ( $k = 1$ :  $2 < 4$ ), the square  $\text{conv}\{(\pm 1, 0), (0, \pm 1)\}$  ( $k = 2$ :  $4 < 8$ ; its 8 faces with antipodal-free vertex sets are the 4 vertices and the 4 edges), and the octahedron ( $k = 3$ :  $6 < 16$ ; all 26 antipodal-free vertex sets are faces: 6 vertices, 12 edges, 8 triangles). For  $k = 0$  the only polytope is a point, with  $1 \neq 2^1$  vertex.

*Remark 3* (Other readings). 1. *Center.* The cross-polytope is symmetric about the origin, so it is centrally symmetric about some center under any convention.

2. *k as the neighborliness parameter.* If “ $k$ -dimensional” is read as “cs- $k$ -neighborly” in arbitrary dimension, then  $C_k^\Delta$  is a cs- $k$ -neighborly polytope with  $2k < 2^{k+1}$  vertices.
3. *Lattice vs. real polytopes.* Our example has integer vertices. A counterexample in a subclass is a counterexample to the universally quantified claim.
4. *Facets instead of vertices.* This is not what is written, but it also fails.  $C_k^\Delta$  has  $2^k < 2^{k+1}$  facets. Under the literal definition, the cube  $[-1, 1]^k$  qualifies and has  $2k$  facets.

## 5 The Chinese reading: dimension $2k$

The Chinese text reads: the minimal number of vertices of a  $2k$ -dimensional cs-neighborly polytope is  $2^{k+1}$ . As a universal statement, this is

$$\forall k \geq 1 : \min\{f_0(P) : P \text{ a } 2k\text{-dimensional cs-neighborly polytope}\} = 2^{k+1}, \quad (2)$$

and in particular it contains the universal lower bound: every  $2k$ -dimensional cs-neighborly polytope has at least  $2^{k+1}$  vertices.

**Theorem 4.** *Claim (2) holds for  $k = 1, 2$  and fails for every  $k \geq 3$ , under each reading of “cs-neighborly”. The smallest counterexample is the 6-dimensional cross-polytope  $C_6^\Delta$ , which has property (F) and  $12 < 16 = 2^4$  vertices.*

*Proof.* By Proposition 5 (below), the minimum in dimension  $2k$  is  $4k$ , under every reading. We have  $4k = 2^{k+1}$  for  $k = 1, 2$  ( $4 = 4$ ,  $8 = 8$ ), and  $4k < 2^{k+1}$  for  $k \geq 3$  ( $12 < 16$ , and if  $4k < 2^{k+1}$  then  $4(k+1) = 4k + 4 < 2^{k+1} + 2^{k+1} = 2^{k+2}$ , using  $4 \leq 2^{k+1}$ ). For  $k \geq 3$ ,  $C_{2k}^\Delta$  is a counterexample (Theorem 1).  $\square$

So the Chinese version is also false as a statement about all  $k$ , but its first two instances are true; a counterexample needs dimension at least 6. Lean proves `conjecture_00000001564_false_2k`: for every notion  $N$  that  $C_6^\Delta$  satisfies,  $\neg \forall k \geq 1, \text{MinVerticesIs } N(2*k) (2^{k+1})$ , with instances for the literal definition, property (F), cs- $j$ -neighborliness for every  $j$ , and cs-3-neighborliness (Grünbaum’s cs- $\lfloor d/2 \rfloor$ -neighborliness for  $d = 6$ ). The case  $k = 3$  needs only the lower-bound half, which  $C_6^\Delta$  violates. The truth of the instances  $k = 1, 2$  rests on Proposition 5, which is not formalized in dimensions 2 and 4.

## 6 The true minimum and the Hanner clause

**Proposition 5.** *Every  $k$ -dimensional centrally symmetric polytope has at least  $2k$  vertices. Equality holds exactly for the affine images of  $C_k^\Delta$ , and these have property (F). Hence, under each reading above, the minimal number of vertices of a  $k$ -dimensional cs-neighborly polytope is  $2k$ , attained exactly by the affine cross-polytopes.*

*Proof.* Translate the center to 0, so  $V = -V$ . Since  $0 = \frac{1}{2}(v + (-v))$  lies in the affine hull, the affine hull equals the linear span of  $V$ , which is therefore  $\mathbb{R}^k$ . Choose a basis  $v_1, \dots, v_k \in V$ . The  $2k$  points  $\pm v_i$  lie in  $V$  and are distinct:  $v_i \neq 0$ , and  $v_i = \pm v_j$  for  $i \neq j$  contradicts independence. If  $|V| = 2k$ , then  $V = \{\pm v_i\} = AV_k$  with  $A$  the invertible linear map  $e_i \mapsto v_i$ . An invertible affine map preserves faces, central symmetry and antipodality, so  $AC_k^\Delta$  has property (F). Conversely, every affine image of  $C_k^\Delta$  has  $2k$  vertices.  $\square$

*Remark 6* (A “maximal” reading). If “minimal” were a slip for “maximal”, the claim would still fail for the  $k$ -dimensional reading. Under the literal definition there is no maximum (centrally symmetric  $2m$ -gons exist for every  $m$ ). Under Grünbaum’s cs-2-neighborliness, Linial and Novik [3] proved that a cs-2-neighborly  $d$ -polytope has at most  $2^d$  vertices, so  $2^{k+1}$  is not attained. This remark is not formalized.

**Hanner polytopes.** Hanner polytopes are generated from the segment  $[-1, 1]$  by Cartesian products  $P \times Q$  and free sums  $P \oplus Q = \text{conv}(P \times \{0\} \cup \{0\} \times Q)$ . For centrally symmetric  $P, Q$  with 0 in the interior,  $f_0(P \times Q) = f_0(P)f_0(Q)$  and  $f_0(P \oplus Q) = f_0(P) + f_0(Q)$ , and dimensions add. By induction, a  $d$ -dimensional Hanner polytope satisfies  $2d \leq f_0 \leq 2^d$ . The base case is the segment,  $f_0 = 2$ . For the product step,  $d_1 + d_2 \leq 2d_1d_2$  and  $2^{d_1}2^{d_2} = 2^{d_1+d_2}$ . For the free-sum step,  $x + y \leq xy$  for  $x, y \geq 2$  gives  $2^{d_1} + 2^{d_2} \leq 2^{d_1+d_2}$ . So no Hanner polytope has  $2^{d+1}$  vertices, and the conjecture’s two clauses cannot both hold if “extremal bodies” means the bodies attaining the minimum. The cube ( $2^d$  vertices) and the cross-polytope ( $2d$  vertices) are Hanner. By Proposition 5, the true minimizers are affine cross-polytopes, which are Hanner up to linear equivalence. So the corrected statement “minimum  $2k$ , attained by (affine) Hanner polytopes” is true. (The conjecture may be a garbled form of Kalai’s  $3^d$  conjecture on the total number of faces of cs polytopes, where Hanner polytopes are the conjectured extremal bodies. That is a different statement.)

## 7 Lean formalization

The project `lean4/` (Lean 4.19.0, core library only, no Mathlib) defines all objects from scratch.

- Points of  $\mathbb{Z}^k$  are lists of  $k$  integers, with the scalar product `dot`. A polytope is given by its vertex list `V`.
  - `IsVertex V v`:  $v$  is the unique maximizer over  $V$  of some functional.
  - `ConvexPosition V`: no repetitions, and every point is a vertex, so the polytope has `V.length` vertices.
  - `IsFace V S`:  $S$  is exactly the set of maximizers of some functional.
  - `CentSymm V`:  $V = -V$ .
- `FullDim k V` is a dimension certificate: points  $a, b_1, \dots, b_k \in V$  and functionals  $f_i$  with  $f_i(b_j - a) \neq 0 \iff i = j$ . `fullDim_linIndep` and `fullDim_sound` prove that the  $b_j - a$  are then linearly independent (the same argument works over  $\mathbb{R}$ ). Together with all points lying in  $\mathbb{Z}^k$ , this gives `IsPolytopeOfDim k V`.
- The three readings are `StatementCsNeighborly` (the literal definition), `CsNeighborlyUpTo j` (Grünbaum’s cs- $j$ -neighborliness) and `CsNeighborlyFull` (property (F)). `full_upTo` shows that (F) implies the second for every  $j$ , and `statement_def_of_cs` shows that every nonempty cs polytope satisfies the first.
- `MinVerticesIs N k n` says that the minimum is attained and is a lower bound; `MinClause N` is the clause  $\forall k \geq 1, \text{MinVerticesIs } N \text{ } k \text{ } (2^{k+1})$ .
- `crossFacts_1` to `crossFacts_4` and `crossFacts_6` check, by `decide` for the generated list `cross k`:
  - length  $2k$ , no repetitions, central symmetry;

- the vertex property (witness  $c = v$ ) and the dimension certificate;
- for every subset  $S$  (every sublist), that if  $S$  is nonempty and antipodal-free, then the maximizers of  $\sum S$  are exactly  $S$ .

`full_of_sublists` transfers the last check from sublists to arbitrary subsets.

- Main theorems:

- `conjecture_00000001564_false` :  $\neg \text{MinClause StatementCsNeighborly}$ ;
- `conjecture_00000001564_false_grunbaum` ( $j$ ), for every  $j$ ;
- `conjecture_00000001564_false_full`;
- `conjecture_00000001564_false_any`, for any notion satisfied by the octahedron;
- `conjecture_00000001564_false_param`, the reading in Remark 3(2);
- `conjecture_00000001564_false_2k`: the Chinese  $2k$ -dimensional reading (Section 5), via `crossFacts_6`, with four instances (suffixes `statement`, `full`, `grunbaum`, `cs3`);
- `fails_each_k`: failure at each  $k = 1, 2, 3, 4$ ;
- `conjecture_00000001564_conjunction_false`: the conjunction with any Hanner clause is false.

- Non-vacuity:

- `true_min_k1` proves a true instance, `MinVerticesIs`  $\leq 1/2$ , including its lower bound.
- A centrally symmetric hexagon is a 2-polytope satisfying the literal definition but not cs-2-neighborliness (`hexagon_not_csNeighborly`).
- A set with an interior point is not in convex position.

- `HannerCount` records the (dimension, vertex number) pairs generated by segments, products and free sums. `hanner_bounds` proves  $2d \leq v \leq 2^d$ , and `no_hanner_with_2pow_succ` proves  $v \neq 2^{d+1}$ .

There is no `sorry`, `native_decide` or added axiom. `#print axioms` reports only `propext`, `Quot.sound` and (for the generic linear-independence lemma) `Classical.choice`.

### Scope.

- Lean checks the cross-polytope for  $k = 1, 2, 3, 4, 6$ ; one  $k$  suffices to refute the clause. The case of general  $k$  is Theorems 1 and 2 above, and `verify.py` checks  $k \leq 6$ .
- The general lower bound  $2k$  of Proposition 5 is proved here. It is formalized only for  $k = 1$ , and `verify.py` checks it for lattice sets in  $\{-1, 0, 1\}^k$ ,  $k = 2, 3$ .
- For Hanner polytopes, only the vertex-count recursion is formalized. The facts  $f_0(P \times Q) = f_0(P)f_0(Q)$  and  $f_0(P \oplus Q) = f_0(P) + f_0(Q)$  are standard, and `verify.py` confirms them on explicit point sets.

The integer functionals used as witnesses are real functionals, so every Lean predicate implies its real counterpart. In particular, `conv(cross 3)` is a genuine 3-dimensional cs polytope in  $\mathbb{R}^3$  with 6 vertices and property (F).

## 8 Independent check (verify.py)

The Python script uses only the standard library and a method different from the Lean proof.

- For  $k \leq 6$ , it enumerates all functionals  $c \in \{-1, 0, 1\}^k \setminus \{0\}$ , and for  $k \leq 3$  also all  $c \in \{-2, \dots, 2\}^k \setminus \{0\}$ . It collects every maximizer set and confirms that this family is exactly the family of antipodal-free vertex subsets (2, 8, 26, 80, 242, 728 for  $k = 1, \dots, 6$ ). It also computes the dimension as an exact rank.
- It checks every centrally symmetric point set in  $\{-1, 0, 1\}^k$  ( $k = 2, 3$ ) that spans  $\mathbb{R}^k$ : each gives a polytope with at least  $2k$  vertices.
- It builds all 1, 2, 6, 22, 90, 394 explicit Hanner polytopes of dimension  $1, \dots, 6$  and confirms that their vertex numbers lie in  $[2d, 2^d]$  and never equal  $2^{d+1}$ .

## 9 Reproduction

```
cd lean4 && lake build          # Lean 4.19.0, core only
cd .. && python3 verify.py      # Python 3 standard library
pdflatex report.tex && pdflatex report.tex && pdflatex report.tex
```

## References

- [1] B. Grünbaum, *Convex Polytopes*, 2nd ed., Springer GTM 221, 2003 (chapter on centrally symmetric polytopes).
- [2] O. Hanner, Intersections of translates of convex bodies, *Math. Scand.* 4 (1956), 65–87.
- [3] N. Linial and I. Novik, How neighborly can a centrally symmetric polytope be?, *Discrete Comput. Geom.* 36 (2006), 273–281.
- [4] G. Kalai, The number of faces of centrally-symmetric polytopes, *Graphs Combin.* 5 (1989), 389–391.